8.3.4.5 Hexadecimal Number System

Number	8 A B 5 . C 9
Weight of each digit	16^3 16^2 16^1 16^0 . 16^{-1} 16^{-2}

Which electronic component can work as a binary switch?

The transistor can work as a binary switch.

8.4 CONVERSIONS

Conversion of binary number from one number format to another number format can be performed by adapting some rules and regulations. Some of the important conversion processes are explained below. For the conversion of integer and fractional number, separate conversion methods are used.

8.4.1 Decimal to Binary Conversion

In this case, the decimal number is divided by 2, and writing down the remainder after each division. The remainders are taken in reverse order to form the binary number.

Example: Conversion of 26_{10} to its equivalent binary number

2|26
2|13 – 0
2|6 – 1
2|3 – 0
1 – 1

Hence, $11010_2 = 26_{10}$

8.4.2 Binary to Decimal Conversion

To convert binary number to its equivalent decimal number, multiply each binary digit by its weight and then add the resulting products.

Example: Conversion of 1101_2 to its equivalent decimal number.

1	0	1	1
2^3	2^2	2^1	2^0

Equivalent decimal number

$= (1 \times 2^3) + (0 \times 2^2) + (1 \times 2^1) + (1 \times 2^0)$

$= (1 \times 8) + (0 \times 4) + (1 \times 2) + (1 \times 1)$

$= 8 + 0 + 2 + 1 = 11$

Hence, $1011_2 = 11_{10}$

8.4.3 Decimal to Octal Conversion

In the case of decimal to octal conversion, the decimal number is divided by 8, and writes down the remainder after each division. The remainders are taken in reverse order to form the octal number.

Example: Conversion of the decimal number 408 to its equivalent octal number.

8|408
8|51 – 0
6 – 3

Hence, $408_{10} = 630_8$

8.4.4 Octal to Decimal Conversion

To convert an octal number to its equivalent decimal number, multiply each octal digit by its weight and then add the resulting products.

Example: Conversion of an octal number 375 into its equivalent decimal number.

The weight of 5 is 8^0, 7 is 8^1 and 3 is 8^2.

Hence, the equivalent decimal number is

$= (3x8^2)+(7x8^1)+(5x8^0)$

$= (3x64)+(7x8)+(5x1)$

$= 192+56+5= 253$

Hence, $375_8 = 253_{10}$

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